The Dimension of Nonterminating Computations

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Abstract

How random can an individual random tape be if a computation runs forever on it? We answer this question for rejection and resampling algorithms. For rejection sampling from a computable finite source π, with $0 < \pi(x) < 1$ and nonempty rejection set U, let δ solve $\sum_{x \in U} \pi(x)^\delta = 1$. Every tape rejected forever has effective strong dimension at most δ; some tapes attain the bound, and the nonterminating set has source-relative Hausdorff dimension δ. For general resampling computations, reading a run backward turns the information revealed by repaired events, minus the description length of the repair history, into an exact likelihood ratio. This identity gives an exponential running-time tail when the difference grows linearly and, under computability, a Kolmogorov-complexity bound for every tape. Nested rejection shows why a dependency graph can overcount overlapping events and lose the dimension bound.

1 Introduction

A randomized algorithm maps its random tape into a trajectory. Once the tape is fixed, the computation is deterministic. It either stops or it does not. We study the set

$$\mathcal{N} = \{\omega : T(\omega) = \infty\}.$$

Throughout, T is the number of rejection or repair steps performed before stopping, with $T = \infty$ on a nonterminating tape. The tail probability $\mathbb{P}\{T \geq n\}$ measures how often the computation survives n steps. We ask a different question: how incompressible can a single tape in $\mathcal{N}$ be?

1.1 One tape and the set of all tapes

A familiar example is the simplest form of rejection sampling. Draw $X_1, X_2, \ldots$ independently from a full-support probability law π on a finite set Ω with $|\Omega| \geq 2$, reject a draw in $U \subseteq \Omega$, and return the first draw in $G = \Omega \setminus U$. This is the zero–one form of the classical accept–reject method (von Neumann, 1951; Devroye, 1986).

Write $p = \pi(U)$ and $a = \pi(G) = 1 - p$. If $0 < a < 1$, then

$$\mathbb{P}\{T \geq n\} = p^n, \qquad \mathbb{P}\{T = t\} = p^t a, \qquad \mathbb{E}T = \frac{p}{a}, \qquad \mathbb{E}(T+1) = \frac{1}{a} \qquad (n, t \in \mathbb{Z}_{\geq 0}). \tag{1}$$

Here T is the number of rejected draws and $T + 1$ is the total number of draws. For $a > 0$, the returned value has the conditional law

$$\mathbb{P}\{X_{T+1} \in B\} = \frac{\pi(B \cap G)}{\pi(G)} \qquad (B \subseteq \Omega). \tag{2}$$

If $a = 1$, the first draw is accepted; if $a = 0$, the algorithm never stops. When $0 < a < 1$, the algorithm stops almost surely although the set $U^{\mathbb{N}}$ of tapes on which it never stops is nonempty. Probability assigns this set mass zero. Dimension asks how much randomness can nevertheless remain in one of its elements.

Now suppose that the source is uniform and $U \neq \varnothing$. Write $Q = |\Omega| \geq 2$ and $M = |U|$, so $1 \leq M \leq Q$. If E_n is the event that the first n blocks are rejected, then

$$\mathbb{P}(E_n) = \left(\frac{M}{Q}\right)^n,$$

and there are only M^n possible surviving prefixes. Each such prefix can be described using $n \log_2 M + o(n)$ bits, whereas the source supplies $n \log_2 Q$ bits. Their ratio is

$$\frac{\log M}{\log Q}.$$

In the two-bit example

$$\Omega = \{00, 01, 10, 11\}, \qquad U = \{00, 01\}.$$

Every surviving block has its first bit fixed and its second bit free. Exactly one half of the source information can remain.

For a general source $P = \pi^{\mathbb{N}}$, define the information in a prefix by

$$I_P(\omega_{1:n}) = -\log_2 P[\omega_{1:n}].$$

Let $K(w)$ be the length of the shortest prefix-free program that prints w. The effective strong dimension of a tape is

$$\mathrm{Dim}_{\mathrm{eff},P}(\omega) = \limsup_{n\to\infty} \frac{K(\omega_{1:n})}{I_P(\omega_{1:n})}. \tag{3}$$

A value near one means that the tape is nearly as incompressible as an ordinary draw from P. A smaller value means that continued execution has imposed structure on the tape.

There are three different questions:

(i) $\mathbb{P}(E_n)$ is the probability of running for at least n steps;

(ii) $\mathrm{Dim}_{\mathrm{eff},P}(\omega)$ measures the randomness of one tape; and

(iii) $\dim_H^P \mathcal{N}$ measures the size of the set of all nonterminating tapes.

For rejection sampling, the tail in (i) is $\pi(U)^n$. The two dimension questions are governed by the number δ that solves

$$\sum_{x\in U} \pi(x)^\delta = 1. \tag{4}$$

Then every tape rejected forever satisfies

$$\mathrm{Dim}_{\mathrm{eff},P}(\omega) \leq \delta \qquad (\omega \in U^{\mathbb{N}}). \tag{5}$$

Moreover,

$$Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}.$$

This is a probability law, and a tape that is Martin–Löf random under $Q_\delta^{\mathbb{N}}$ attains equality. The classical set calculation gives $\dim_H^P(U^{\mathbb{N}}) = \delta$. Under the uniform law, $\pi(U) = Q^{\delta-1}$, so the tail exponent and the dimension determine one another. For a nonuniform law, they need not.

The effective-dimension statement is pointwise: it concerns each tape separately. Hausdorff dimension measures the whole nonterminating set and shows that the pointwise bound is sharp.

1.2 Reconstructing the random tape

Now consider a resampling computation. At each step it selects a bad event and redraws the variables on which that event depends. A finite run can be read backward if we know its final assignment, the ordered list of repairs, and the values erased by those repairs. Starting from the final assignment, we restore the erased values in reverse order and recover every random choice used by the run.

This reconstruction gives a simple comparison:

information revealed by the repaired events
− information needed to describe the repair history = a log-likelihood ratio.

Because the backward descriptions have total probability at most one, this ratio has an exponential upper tail. If the description is computable, it also gives a program for the used part of the random tape: the tape can be shorter than its source description by at least the same number of bits. Theorem 2.1 states both conclusions exactly. Event probabilities alone are not enough; the description must include the repair order and any control information needed to reverse the run.

1.3 Nested rejection rules

A dependency graph has one vertex for each rejection rule and joins two rules that may interact. In the variable model, two rules are joined when they depend on a common random variable. The Lovász local lemma and its variants use this graph to give conditions under which all bad events can be avoided (Erdős and Lovász, 1975; Shearer, 1985; Scott and Sokal, 2005). Algorithmic analyses use the same graph to bound the number and weight of repair histories (Moser and Tardos, 2010; Haeupler and Harris, 2017; Fialho et al., 2020). The graph does not record how the rejected sets intersect or whether one contains another.

Suppose several nested rules can reject the same random block. The algorithm records the smallest rule that applies and then redraws the whole block. Its true rejection probability is

$$\pi\left(\bigcup_A A\right),$$

whereas a dependency-graph calculation treats the rules separately and uses $\sum_A \pi(A)$. The latter can count one outcome several times. Dividing the union into disjoint cases preserves the recorded rule and recovers the exact rate. Section 3 proves this statement and gives an example with nested Hamming balls in which the graph calculation loses the dimension bound entirely.

Because every rejection redraws the whole block, this example has no persistent state. Local repairs, which retain some variables, are the harder case.

1.4 Contribution and limits

Theorem 2.1 is the main result. It expresses a finite run as an exact likelihood ratio. Applying a probability bound to this ratio controls the running time; coding the same backward law controls the Kolmogorov complexity of each tape. Nested rejection then identifies precisely where a dependency graph overcounts.

The finite-state relation between weighted growth, Hausdorff dimension, and the corresponding tilted Markov law is classical (Parry, 1964; Mauldin and Williams, 1988; Bowen, 1975; Simpson, 2015). Reversible records are fundamental to entropy compression, and their Kolmogorov-complexity interpretation is also known (Moser, 2009; Moser and Tardos, 2010; Fortnow, 2009). We use these results rather than restate them as new.

Recent work studies complexity at a realized network, a selected Gaussian index, or the state needed for a Bellman recursion (Li and Xu, 2026; Xu, 2026a,b). Here the realized object is the random tape. The present results do not imply a learning-theoretic generalization bound.

The unresolved case is local repair. When two repairs act on overlapping sets of variables, values outside the refreshed set persist. A rule label alone need not determine the next step. One must retain enough boundary state to compute the rate of continued execution and still reverse the run.

2 Reversing a resampling computation

Let $X_1, \dots, X_n$ be independent finite-valued variables with product law $\pi = \bigotimes_i \pi_i$. Each bad event A is determined by coordinates $v(A)$ and has probability p_A. Fix a deterministic rule for selecting a currently true bad event.

We list the random choices in the order in which the algorithm uses them. First draw $S_0 \sim \pi$. Whenever the current assignment contains a bad event, select A_t, draw $Z_t \sim \pi_{v(A_t)}$ independently of the past, and write Z_t on $v(A_t)$. Let E_T be the event that at least T resamplings occur, and put

$$U_T = (S_0, Z_1, \dots, Z_T).$$

The tape prefixes in E_T are disjoint, and their total probability is $\mathbb{P}(E_T)$.

Let Y_t be the configuration overwritten at time t, let S_T be the assignment after T repairs, and let $L_T = (A_1, \dots, A_T)$. Coordinatewise, the entries of the used tape are partitioned into the final assignment and the overwritten configurations. Therefore

$$P_{\text{tape}}(U_T) = \pi(S_T) \prod_{t=1}^{T} p_{A_t}\, \pi_{v(A_t)}(Y_t \mid A_t). \tag{6}$$

Let Q_T assign nonnegative weights of total mass at most one to possible lists of T selected events. Define a comparison law on complete records by

$$G_T(s, \ell, y_{1:T}) = \pi(s) Q_T(\ell) \prod_{t=1}^{T} \pi_{v(A_t)}(y_t \mid A_t). \tag{7}$$

The map

$$U_T \longmapsto (S_T, L_T, Y_{1:T})$$

is injective. Indeed, reverse the repairs from T to 1: the current values on $v(A_t)$ recover Z_t, and restoring Y_t recovers the preceding assignment.

Theorem 2.1 (The information identity)**.** *Every execution in E_T satisfies*

$$\log_2 \frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(U_T)} = \sum_{t=1}^{T} \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(L_T)}. \tag{8}$$

Call the right side Δ_T on E_T and set $\Delta_T = -\infty$ otherwise. Then, for every $u \geq 0$,

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}. \tag{9}$$

If the instance and Q_T are computable, then

$$-\log_2 P_{\text{tape}}(U_T) - K(U_T \mid T, \mathit{instance}) \geq \Delta_T - O(1). \tag{10}$$

Proof. Divide (7) by (6). The final-state and conditional overwritten-value factors cancel, giving (8). Pull G_T back through the injective record map. This gives weights τ_T of total mass at most one on live tape prefixes, with $\tau_T(U_T)/P_{\text{tape}}(U_T) = 2^{\Delta_T}$. Hence

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u} \sum_{U_T} \tau_T(U_T) \leq 2^{-u}.$$

If τ_T is computable, Shannon–Fano coding (Shannon, 1948; Fano, 1949) gives $K(U_T \mid T, \text{instance}) \leq -\log_2 \tau_T(U_T) + O(1)$, which rearranges to (10). $\square$

The two terms in (8) have different origins. The first is the information revealed by the events that were repaired. The second is the number of bits needed to describe which repairs occurred. Neither term alone gives a bound; their difference does, because it is a likelihood ratio.

For example, color the vertices of a graph independently with q colors and repeatedly recolor the endpoints of the first monochromatic edge. A fixed edge is monochromatic with probability $1/q$, so each repair contributes $\log_2 q$ bits to the first term. The second term pays for the repaired edges and their order. The comparison is therefore between information ruled out by the violations and information used by the algorithm's history.

Corollary 2.2 (A tail bound and a dimension bound)**.** *Suppose every T-repair execution satisfies*

$$\sum_{t=1}^{T} \log_2 \frac{1}{p_{A_t}} \geq \beta T - C_0, \qquad -\log_2 Q_T(L_T) \leq \gamma T + C_1,$$

where $\beta > \gamma$. Then

$$\mathbb{P}(E_T) \leq 2^{C_0 + C_1 - (\beta - \gamma)T}. \tag{11}$$

For the pointwise conclusion, assume in addition that

(i) the instance and the family $(Q_T)_{T\geq 1}$ are uniformly computable;

(ii) each repair consumes exactly c fair source bits; and

(iii) the repair-depth prefixes are nested in a fixed computable tape, with sublinear interpolation overhead.

Then every nonterminating tape satisfies

$$\mathrm{Dim}_{\mathrm{eff}}(\omega) \leq \max\left\{0, 1 - \frac{\beta - \gamma}{c}\right\}. \tag{12}$$

Proof. The assumptions give $\Delta_T \geq (\beta - \gamma)T - (C_0 + C_1)$. Equation (11) follows from (9). Equation (10) gives $K(U_T) \leq cT - (\beta - \gamma)T + o(T)$ along an infinite execution. Divide by cT and interpolate between consecutive repair prefixes. $\square$

Remark 2.3 (What the record must contain). For a recursive Moser-style algorithm, the record must include the recursion tree or return structure, not only the clause labels. Any argument that omits this information must supply another prefix-free decoding rule. In Theorem 2.1, everything needed for reversal contributes to the probability $Q_T(L_T)$.

3 What a dependency graph forgets

To use Theorem 2.1, one must describe the possible lists of repairs. The Lovász local lemma summarizes event interactions by a dependency graph, and its algorithmic proofs use that graph to organize repair histories.

3.1 Why dependency graphs enter

Let $\mathcal{A}$ be a finite family of bad events. The Lovász local lemma asks when there is a positive probability of avoiding all of them:

$$\mathbb{P}\left(\bigcap_{A\in\mathcal{A}} A^c\right) > 0. \tag{13}$$

In the variable model, where independent variables generate the events, the standard graph D joins A and B whenever their variable sets overlap. For example, in random vertex coloring a bad event says that one edge is monochromatic, and two such events are adjacent in D when their edges share a vertex. More generally, write $\Gamma(A)$ for the neighbors of A. A loopless graph on $\mathcal{A}$ is a dependency graph if every event A is independent of $\sigma(B : B \in \mathcal{A} \setminus (\{A\} \cup \Gamma(A)))$. The vertices of D are events; this is different from the bipartite variable–event incidence graph used in Section 3.5. If $\mathbb{P}(A) \leq p$ for every A and D has maximum degree d, the symmetric local lemma gives the familiar sufficient condition

$$\mathrm{e}\, p(d+1) \leq 1. \tag{14}$$

More generally, Shearer's region gives the sharp worst-case criterion based only on a graph and the marginal event probabilities (Erdős and Lovász, 1975; Shearer, 1985; Scott and Sokal, 2005).

Moser and Tardos turn this existence argument into a random process. Draw all variables independently; while some bad event is true, choose one and redraw the variables on which it depends. If there are numbers $x_A \in (0,1)$ such that

$$\mathbb{P}(A) \leq x_A \prod_{B\in\Gamma(A)} (1 - x_B) \qquad (A \in \mathcal{A}), \tag{15}$$

then the expected number of times A is redrawn is at most $x_A/(1 - x_A)$ (Moser and Tardos, 2010). Consequently,

$$\mathbb{E}T \leq \sum_{A\in\mathcal{A}} \frac{x_A}{1 - x_A} < \infty. \tag{16}$$

The algorithm therefore stops almost surely and returns an assignment in which no bad event occurs. Witness trees, witness DAGs, and entropy-compression records prove bounds of this kind by encoding repair histories through event probabilities and adjacency in D (Moser and Tardos, 2010; Haeupler and Harris, 2017; Fialho et al., 2020).

This economy has a cost. The graph records which repairs may interact, but it does not determine $\mathbb{P}(A \cap B)$, set inclusion, or $\mathbb{P}(\bigcup_A A)$. Two event systems can therefore have the same graph and the same marginal probabilities but different stopping rates and different nonterminating sets. The next three subsections compare the exact state-space description, the graph bound, and a nested example where the loss is explicit.

3.2 The full state space

Every finite variable-model resampling algorithm is a finite-state randomized computation on assignments. Fix a deterministic rule for selecting a currently true bad event. On reachable nonterminal assignments define

$$(B_\theta)_{xy} = \sum_{z\colon F(x,z)=y} \pi_{A(x)}(z)^\theta, \tag{17}$$

where $A(x)$ is the selected event and $F(x,z)$ is the updated assignment. Decompose the nonterminal transition graph into reachable strongly connected components C containing a directed cycle. Proposition A.1 gives the exact source-relative dimension

$$\dim_H^P \mathcal{N} = \max_C \delta_C, \qquad \rho(B_{\delta_C,C}) = 1. \tag{18}$$

This formula is exact, but its matrix usually has exponentially many rows and columns.

3.3 The graph bound

Classical entropy-compression and Shearer arguments use the dependency graph and the probabilities p_A to bound the total weight of possible repair lists (Cartier and Foata, 1969; Scott and Sokal, 2005). We need only the complete-graph case. If $D = K_m$, the graph-only enumeration allows every word $(A_1, \ldots, A_t) \in \mathcal{A}^t$ and assigns it weight $\prod_{i=1}^t p_{A_i}$. Its total weight is

$$\left(\sum_A p_A\right)^t.$$

This is an upper enumeration based only on D and the marginals; it need not equal the probability mass of the histories produced by a fixed selection rule. For uniform q-ary blocks of size b, this gives the dimension bound in (29). The bound is classical. Moreover, a short graph description does not by itself give an efficient calculation: the relevant independence polynomials and the general Shearer region are not polynomial-time computable in general (Harvey et al., 2018).

3.4 Nested rejection rules

Let Ω be a finite block space with source law π satisfying $0 < \pi(x) < 1$ for every x, and let $\mathcal{A}$ be a laminar family of bad subsets: for $A, B \in \mathcal{A}$, either $A \cap B = \varnothing$, $A \subseteq B$, or $B \subseteq A$. Every repair resamples the whole block from π. If the current block x lies in the bad union $U = \bigcup_{A \in \mathcal{A}} A$, select the inclusion-minimal event containing x. This event is unique by laminarity.

Let $\mathrm{ch}(A)$ be the maximal proper members of $\mathcal{A}$ contained in A, and define the disjoint region

$$C_A = A \setminus \bigcup_{B \in \mathrm{ch}(A)} B. \tag{19}$$

Discard empty regions and let $\mathcal{I}$ index the remaining ones. They are disjoint, their union is U, and the algorithm records A exactly on C_A. Define

$$w_A(\theta) = \sum_{x \in C_A} \pi(x)^\theta, \qquad W(\theta) = \sum_{A \in \mathcal{I}} w_A(\theta) = \sum_{x \in U} \pi(x)^\theta. \tag{20}$$

Theorem 3.1 (Hierarchical rejection). *Assume $U \neq \varnothing$, and let $P = \pi^{\mathbb{N}}$. There is a unique $\delta \in [0,1]$ such that*

$$W(\delta) = \sum_{x \in U} \pi(x)^{\delta} = 1. \tag{21}$$

(a) If the alphabet, π, U, and the law Q_δ below are computable, every nonterminating tape satisfies

$$\mathrm{Dim}_{\mathrm{eff},P}(\omega) \leq \delta \qquad (\omega \in \mathcal{N} = U^{\mathbb{N}}). \tag{22}$$

Every tape that is Martin–Löf random for $Q_\delta^{\mathbb{N}}$ satisfies $\dim_{\mathrm{eff},P}(\omega) = \mathrm{Dim}_{\mathrm{eff},P}(\omega) = \delta$. Moreover,

$$\dim_H^P \mathcal{N} = \delta, \qquad \mathbb{P}\{T \geq t\} = \pi(U)^t. \tag{23}$$

Under the uniform law on $Q = q^b$ blocks, writing $M = |U|$,

$$\dim_H \mathcal{N} = \delta = \frac{\log M}{\log Q}, \qquad \mathbb{P}\{T \geq t\} = \left(\frac{M}{Q}\right)^t. \tag{24}$$

In particular, under a uniform source, a tape of effective strong dimension greater than δ must make the algorithm terminate.

(b) Group rejected blocks according to the region C_A that contains them. For every $\theta \geq 0$, the resulting weighted transition matrix is

$$\widehat{B}_\theta(A,B) = w_B(\theta). \tag{25}$$

Its spectral radius is $W(\theta)$, the same as that of the full transition matrix, and $h \equiv 1$ is a right eigenfunction.

(c) The two laws relevant to dimension and survival can be realized by reversible records. The first is

$$Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}. \tag{26}$$

For every T-repair execution, there is a probability record law $G_{\delta,T}$ satisfying

$$\log_2 \frac{G_{\delta,T}(\mathrm{record})}{P_{\mathrm{tape}}(U_T)} = (1-\delta) \sum_{t=1}^{T} \log_2 \frac{1}{\pi(Y_t)}. \tag{27}$$

The second is the source conditioned on another rejection, $Q_{\mathrm{surv}}(x) = \pi(x)\mathbf{1}\{x \in U\}/\pi(U)$, and it satisfies

$$\log_2 \frac{G_{\mathrm{surv},T}(\mathrm{record})}{P_{\mathrm{tape}}(U_T)} = T \log_2 \frac{1}{\pi(U)}. \tag{28}$$

Proof. For full nonterminal states $x, y \in U$, resampling the common scope gives $B_\theta(x,y) = \pi(y)^\theta$. Hence, for every $x \in C_A$,

$$\sum_{y \in C_B} B_\theta(x,y) = \sum_{y \in C_B} \pi(y)^\theta = w_B(\theta),$$

independently of x. Thus grouping states by the sets C_A leaves the weighted transition calculation unchanged and proves (25). The reduced matrix is $\mathbf{1}w(\theta)^{\mathsf{T}}$, so its only nonzero eigenvalue is $W(\theta)$ and $h \equiv 1$ is a right eigenfunction. Proposition A.1 gives (21). Each repair draws a fresh block from π and continues precisely on U, proving the survival formula and its uniform specialization.

For the dimension bound, observe that Q_δ is computable and supported on U. Every $\omega \in U^{\mathbb{N}}$ therefore satisfies

$$K(\omega_{1:n}) \le -\log_2 Q_\delta^{\mathbb{N}}[\omega_{1:n}] + O(\log n) = \delta\bigl(-\log_2 P[\omega_{1:n}]\bigr) + O(\log n).$$

Division by the source information proves (22). If ω is Martin–Löf random for $Q_\delta^{\mathbb{N}}$, Levin–Schnorr gives the reverse inequality up to $O(1)$, and hence equality. This also proves (5).

It remains to construct the record laws. The repair list is determined by the overwritten states because $A_t = \phi(Y_t)$. Reversal recovers the used tape from $(S_T, Y_{1:T})$. The source factorization is

$$P_{\text{tape}}(U_T) = \pi(S_T) \prod_{t=1}^{T} \pi(Y_t).$$

Since (21) holds, the record law

$$G_{\delta,T}(s, y_{1:T}) = \pi(s) \prod_{t=1}^{T} \Bigl[\pi(y_t)^\delta \mathbf{1}\{y_t \in U\}\Bigr]$$

has total mass one. Dividing by the source factorization proves (27). Replacing each critical factor by $\pi(y_t)\mathbf{1}\{y_t \in U\}/\pi(U)$ gives a probability law and proves (28). Conditioning further on the region C_B recovers the recorded rejection label. □

Remark 3.2 (The one-state reduction). Every repair redraws the whole block. Consequently, the next step is independent of the current rejected block, and the one-state matrix $[W(\theta)]$ already gives the stopping rate. The disjoint regions retain the recorded reason for rejection, but do not change the stopping dynamics. Local repairs, which leave part of the state unchanged, are the harder case.

The survival law is also a direct application of Theorem 2.1. Relabel each repair by the disjoint region C_A containing the overwritten block; this does not change the run or its resampling rule. Put $r_A = \pi(C_A)$ and choose

$$Q_T(A_1, \ldots, A_T) = \prod_{t=1}^{T} \frac{r_{A_t}}{\pi(U)}.$$

Encoding an overwritten block under $\pi(\cdot \mid C_{A_t})$ exposes $\log_2(1/r_{A_t})$ bits, while the log costs $\log_2(\pi(U)/r_{A_t})$ bits. Their difference is exactly $\log_2(1/\pi(U))$ at every repair, giving (28). Conditioning only on the original event A_t would count overlaps; the disjoint regions retain exactly the information needed by the code.

Comparison with the dependency graph. Under the standard variable-overlap definition, all common-block events are pairwise adjacent. The graph calculation therefore uses the one-step weight $\sum_A \pi(A)$, while the true rejection probability is $\pi(U)$. Under a uniform source, the resulting upper bound and the true dimension are

$$1 + \frac{\log \sum_A \pi(A)}{\log Q} \quad \text{and} \quad 1 + \frac{\log \pi(U)}{\log Q}, \tag{29}$$

again truncated to $[0, 1]$. They agree when the events are disjoint and differ when events overlap.

In the variable model, a nonedge certifies disjoint scopes and hence commuting repairs; an edge marks only possible interaction. The complete graph supplies no nonedges, so the missing intersection information is the relevant loss here. The following standard family makes it explicit.

Example 3.3 (Nested Hamming balls). Let $\Omega = \{0,1\}^b$ be uniform and reject a block when its Hamming weight is at most r. Writing

$$U_r = \{x : |x| \le r\}, \qquad M_r = \sum_{j=0}^{r} \binom{b}{j},$$

Theorem 3.1 gives, for every nonterminating tape,

$$\mathrm{Dim}_{\mathrm{eff}}(\omega) \le \frac{1}{b}\log_2 M_r \quad \text{for every } \omega \in U_r^{\mathbb{N}}, \qquad \dim_H(U_r^{\mathbb{N}}) = \frac{1}{b}\log_2 M_r. \tag{30}$$

Now add the stricter rule $U_s \subset U_r$, where $s < r$, and record U_s whenever it applies. The standard variable-overlap dependency graph on the two original rules is K_2 and pays $(M_s + M_r)/2^b$, although the computation rejects on a set of mass only $M_r/2^b$.

For example, take $b \ge 3$, $r = b-1$, and $s = b-2$. The graph sum is $2 - (b+2)/2^b > 1$, so the graph bound gives no information, whereas the true dimension is $b^{-1}\log_2(2^b - 1) < 1$. Splitting U_r into U_s and $U_r \setminus U_s$ removes this double counting.

This does not contradict tightness results for the Shearer region. Those results concern the worst event system with given graph and marginals; Example 3.3 concerns one fixed system whose intersections are known. Related separations in the variable model appear in He et al. (2021).

3.5 Local repairs

The preceding example is simple because every action redraws the same block. After a redraw, the next step depends only on the fresh block. A local repair leaves other variables unchanged, and these retained values can affect the next repair.

Existing witness structures retain selected parts of this information (Achlioptas and Iliopoulos, 2016; Haeupler and Harris, 2017; Fialho et al., 2020). A concrete question is the following. Consider a q-ary resampling system whose variable–event incidence graph has pathwidth w. Can the variables on the boundary of a path decomposition be used as the states of a matrix of size $q^{O(w)}\,\mathrm{poly}(n)$ whose spectral radius gives the true dimension? Can the corresponding tilted law be realized by reversible execution records, giving matching upper and lower bounds for the effective dimension of a tape? Forbidden-pattern generation and bounded-pathwidth k-SAT are natural test cases.

A spectral calculation alone gives no code for actual runs. Conversely, a code that counts histories more than once need not give the true dimension.

4 Conclusion

This paper studies nontermination at the level of one random tape. Its main contribution is an exact backward likelihood identity for resampling computations. The information revealed by repaired events, minus the information needed to describe the repair history, is simultaneously a change-of-measure quantity controlling long runs and, under computability, a lower bound on how many source bits can be saved when describing the tape. Running time and pointwise Kolmogorov complexity are therefore two consequences of the same comparison. Hausdorff dimension plays a different role: it measures the whole nonterminating set and shows when the pointwise bound is sharp.

For rejection sampling and nested rejection rules, this program is exact. The critical law produces tapes attaining the dimension bound, while disjoint event regions show precisely how a dependency-graph enumeration can overcount. The broader goal is a theory that, for a randomized computation, finds a compact state description sufficient to compute the nontermination rate, construct the critical tape law, and reverse individual runs.

Local repair is the main obstacle. Values outside the refreshed coordinates persist, so a repair label or dependency graph may omit state needed to predict and reverse the next step; the full assignment is exact but generally exponential. The open problem is to find an intermediate state description that is neither exponentially large nor information-losing, and to use it to obtain matching bounds for the running-time exponent, individual-tape dimension, and Hausdorff dimension under genuinely overlapping repairs.

A Finite-state computations

Let a finite nonterminal state graph have edges $e : s \to t$ with source probabilities $p(e) > 0$; missing probability mass terminates. For $\theta \geq 0$, define

$$(B_\theta)_{st} = \sum_{e:s\to t} p(e)^\theta, \qquad \Lambda(\theta) = \log \rho(B_\theta). \tag{31}$$

The number $\Lambda(\theta)$ is the exponential growth rate of the total weight of surviving paths; it is usually called pressure. Assume for simplicity that the reachable nonterminal graph is irreducible and that its edge probabilities are bounded away from one. Reducible systems are handled componentwise.

Give a cylinder $[w]$ the source gauge $P[w]$. Its source-relative Hausdorff dimension is the critical d for covers weighted by $P[w]^d$; under a uniform q-ary source this is ordinary Hausdorff dimension in the q-adic metric. The effective lower dimension $\dim_{\mathrm{eff},P}$ is defined as in (3), with $\liminf$ in place of $\limsup$.

Proposition A.1 (The finite-state formula)**.** *Let δ be the unique solution of $\rho(B_\delta) = 1$. Then*

$$\lim_{n\to\infty} \frac{1}{n} \log \sum_{w:\,T(w)\geq n} P[w]^\theta = \Lambda(\theta), \tag{32}$$

$$\lim_{n\to\infty} -\frac{1}{n} \log \mathbb{P}\{T \geq n\} = -\Lambda(1), \tag{33}$$

$$\dim_H^P \mathcal{N} = \delta. \tag{34}$$

If $B_\delta h = h$ with $h > 0$, the equilibrium law

$$Q_\delta(e \mid s) = p(e)^\delta \frac{h_t}{h_s} \tag{35}$$

satisfies, for every surviving path $w = e_1 \cdots e_n$,

$$\log_2 \frac{Q_\delta(w)}{P(w)} = (1-\delta) \log_2 \frac{1}{P[w]} + \log_2 \frac{h_{s_n}}{h_{s_0}}. \tag{36}$$

For a computable system with computable equilibrium law, every nonterminating tape satisfies $\mathrm{Dim}_{\mathrm{eff},P}(\omega) \leq \delta$*. A tape that is Martin–Löf random under Q_δ satisfies* $\dim_{\mathrm{eff},P}(\omega) = \mathrm{Dim}_{\mathrm{eff},P}(\omega) = \delta$*.*

Proof. Equation (32) is the matrix identity $\sum_w P[w]^\theta = e_{s_0}^\mathsf{T} B_\theta^n \mathbf{1}$ followed by Perron–Frobenius theory. Equations (33)–(34) and the Frostman construction are standard graph-directed results (Bowen, 1975; Mauldin and Williams, 1988). Equation (36) follows by telescoping (35). The effective statement is the usual coding and Levin–Schnorr argument (Lutz, 2003; Athreya et al., 2007; Simpson, 2015). ☐

Under a uniform q-ary source, if M_{st} counts symbols moving s to t, then $B_\theta = q^{-\theta} M$ and

$$\dim_H \mathcal{N} = \log_q \rho(M) = 1 - \frac{\text{exponential decay rate in bits}}{\log_2 q}.$$

For a uniform source, the tail exponent determines the dimension by this identity. Equation (36) is the finite-state form of the comparison used in Theorem 2.1.

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